SYNOPSIS

Pre Dissertation

(CSE461)

PROJECT TITLE: -

IADSS: Intelligent Agricultural Decision Support System - An Ensemble Learning Framework for Multi-Domain Agricultural Intelligence

Submitted To

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PROJECT DESCRIPTION

Agriculture plays a vital role in India's economy, yet farmers face challenges such as crop diseases, soil nutrient imbalance, volatile market prices, and limited access to expert advice. Existing agricultural advisory systems are fragmented, requiring farmers to rely on multiple tools, which leads to inefficient and delayed decision-making.

To overcome these limitations, this project proposes an Intelligent Agricultural Decision Support System (IADSS)—a unified, AI-powered platform that integrates plant disease detection, soil analysis, crop price forecasting, and conversational agricultural assistance. The system provides accurate, explainable, and user-friendly recommendations through a single interface.

Problem and Objective

Current solutions rely on isolated machine-learning models, lack transparency, offer poor usability for non-technical users, and cannot adapt to new data. The objective of IADSS is to develop an integrated and adaptive agricultural advisory system using ensemble learning and explainable AI to improve decision quality and accessibility for farmers.

Methodology and Scope

The system employs CNN-based disease detection, regression-based soil analysis, ensemble crop price prediction, and a conversational AI interface, all integrated through a RESTful architecture. It targets farmers, extension services, and agribusinesses, offering real-time analysis, multi-horizon forecasts, and natural language support via a scalable web platform.

Expected Outcomes

IADSS aims to improve prediction accuracy, enhance user satisfaction, reduce technical barriers for farmers, and deliver actionable, expert-validated agricultural recommendations with measurable social and economic impact.

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Module Description:

MODULE 01: Plant Disease Detection System

Overview

This module implements computer vision-based automatic plant disease identification from leaf images, providing farmers with instant diagnostic capabilities previously requiring expert consultation.

Key Features

- Multi-class disease classification supporting 10+ common plant diseases

- Real-time image processing and prediction (< 3 seconds response time)

- Confidence scoring for each prediction

- Disease information including symptoms, causes, and treatment protocols

- Image preprocessing pipeline for handling varying lighting and image quality

. Dataset

- **Source**: Plant Village Dataset + Roboflow Plant Diseases Dataset

- **Classes**: Healthy leaves + 10 disease types (leaf blight, powdery mildew, bacterial spot, etc.)

- **Base Model**: Transfer Learning with Yolo v9.

. Input/Output Specification

Input:

- Image file (JPEG/PNG format)

- Resolution: Minimum 224x224 pixels

- Upload via web interface or API endpoint

Output:

- Disease classification label

- Disease description

- Treatment recommendations

- Preventive measures

. Integration Points

- Connects to Conversational AI module for detailed disease explanations

- Feeds data to analytics dashboard for disease prevalence tracking

- API endpoint: /Api/disease-detection (POST request with image file)

MODULE 02: Soil Analysis and Recommendation

Overview

This module analyses soil nutrient parameters and provides personalized fertilizer recommendations and crop suitability assessments based on NPK values, pH levels, and environmental factors.

Key Features

- Soil nutrient prediction from basic input parameters

- Crop-specific fertilizer recommendations

- Optimal crop suggestions based on soil conditions

- Nutrient deficiency identification

- Season-aware recommendations

. Soil Quality Prediction Model

- **Algorithm**: Random Forest Regression for NPK prediction

- **Features**: Temperature, humidity, rainfall, pH, crop type

- **Target Variables**: Good, Medium, Low (Quality of soil)

- **Performance Metrics**: Accuracy score, Confusion matrix

. Crop Recommendation Engine

- **Approach:** Retrieval-Augmented Generation (RAG) integrated with an LLM
- **Input:** Soil parameters (N, P, K, pH), environmental factors (temperature, humidity, rainfall), and regional context
- **Process:**
    - Structured inputs are first interpreted by the system
    - Relevant agricultural knowledge (crop suitability guides, regional advisories) is retrieved from a vector database
    - The LLM generates recommendations grounded in retrieved documents
- **Output:** Top 3 suitable crops with confidence scores and natural-language explanations
- **Classes:** 20+ crop types (wheat, rice, maize, cotton, etc.)

Fertilizer Recommendation System

- **Approach:** RAG-based LLM with agronomic rule grounding

- **Knowledge Base:** Soil fertility manuals and fertilizer guidelines from verified agricultural sources
- **Logic:**
- IF Nitrogen deficiency detected → Retrieve urea usage guidelines
- IF Phosphorus deficiency detected → Retrieve DAP usage guidelines
- IF Potassium deficiency detected → Retrieve MOP usage guidelines
- **Generation:**
  The LLM produces fertilizer recommendations strictly based on retrieved guidelines, including type, quantity, and method of application
- **Personalization:** Recommendations are adjusted using crop type, growth stage, soil condition, and regional factors

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. Input/Output Specification

Input:

- Soil parameters (N, P, K, pH)
- Environmental data (temperature, humidity, rainfall)
- Optional crop preference
- Location/region

Output:

- Soil quality assessment (Poor / Average / Good / Excellent)
- Nutrient deficiency or excess indicators
- Top 3 crop recommendations with explanations
- Fertilizer type and quantity recommendations
- Optimal fertilizer application timing

6 Integration Points

- Integrates with the crop price prediction module to provide economic insights for LLM-recommended crops
- Connects to the RAG-powered conversational assistant for detailed, context-aware soil and nutrient management queries

MODULE 03: Crop Price Prediction (Ensemble Model)

Overview

This module implements a novel ensemble learning framework combining multiple algorithms to predict crop prices with superior accuracy compared to single-model approaches, enabling farmers to make informed market timing decisions.

Key Features

- Multi-horizon forecasting (7, 14, 30-day predictions)
- Ensemble model combining Random Forest, Prophet, and Meta-Regression
- Confidence intervals for predictions
- Historical price trend visualization
- Market timing recommendations

Technical Implementation

Dataset

- **Source**: Research based data collected and structured by researchers.
- **Duration**: 2+ years of historical price data

2. Ensemble Architecture

A. Random Forest Regressor

- Captures non-linear relationships
- Handles multiple features effectively
- Parameters: x..
- Features: Lagged prices, weather, demand indicators

B. Prophet Time-Series Model

- Specialized for seasonal patterns

C. Meta-Regression Model

- Combines predictions from RF and Prophet

5. Input/Output Specification

Input:

- Crop type

- Current market price

Output:

- Predicted price with confidence interval

- Price trend (increasing/decreasing/stable)

- Volatility indicator

- Market timing recommendation (sell now / wait)

7. Research Contribution

This module forms the core research contribution suitable for publication:

- Novel ensemble architecture for agricultural forecasting

- Comparative analysis demonstrating superiority

- Statistical validation of improvements

MODULE 04: Conversational AI Chatbot

Overview

This module provides an intelligent chatbot interface using a **Retrieval-Augmented Generation (RAG) framework integrated with the Google Gemini API**, enabling accurate, explainable, and context-aware agricultural guidance through natural language interaction. The RAG layer ensures that all responses are grounded in verified agricultural knowledge rather than relying solely on generative reasoning...

Key Features

- Natural language query processing using LLM
- Retrieval-augmented responses grounded in agricultural documents
- Context-aware multi-turn conversations
- Domain-specific agricultural knowledge (disease, soil, fertilizer, market)
- Integration with all system modules
- Personalized recommendations based on user profile and region

1. Prompt Engineering with RAG Framework

Domain-Optimized Prompt Templates:

The chatbot uses **domain-optimized prompt templates** combined with **retrieved knowledge chunks** from a vector database containing verified agricultural documents (crop guides, soil manuals, fertilizer recommendations, market advisories).

Template 1: Disease Explanation Prompt

Template 2: Soil Management Prompt (RAG-Enhanced).

Template 3: Price Advisory Prompt

Template 4: General Agricultural Query

4. Input/Output Specification

Input

- User message
- Conversation history
- User profile data
- System state (disease results, soil analysis, price predictions)

Output

- RAG-grounded AI response
- Suggested follow-up questions
- Relevant module recommendations
- Updated conversation context

5. Integration Points

- Retrieves disease knowledge linked to Module 1 outputs
- Uses soil analysis results from Module 2 for grounded explanations
- Incorporates price predictions from Module 3

MODULE 05: System Integration & API Development

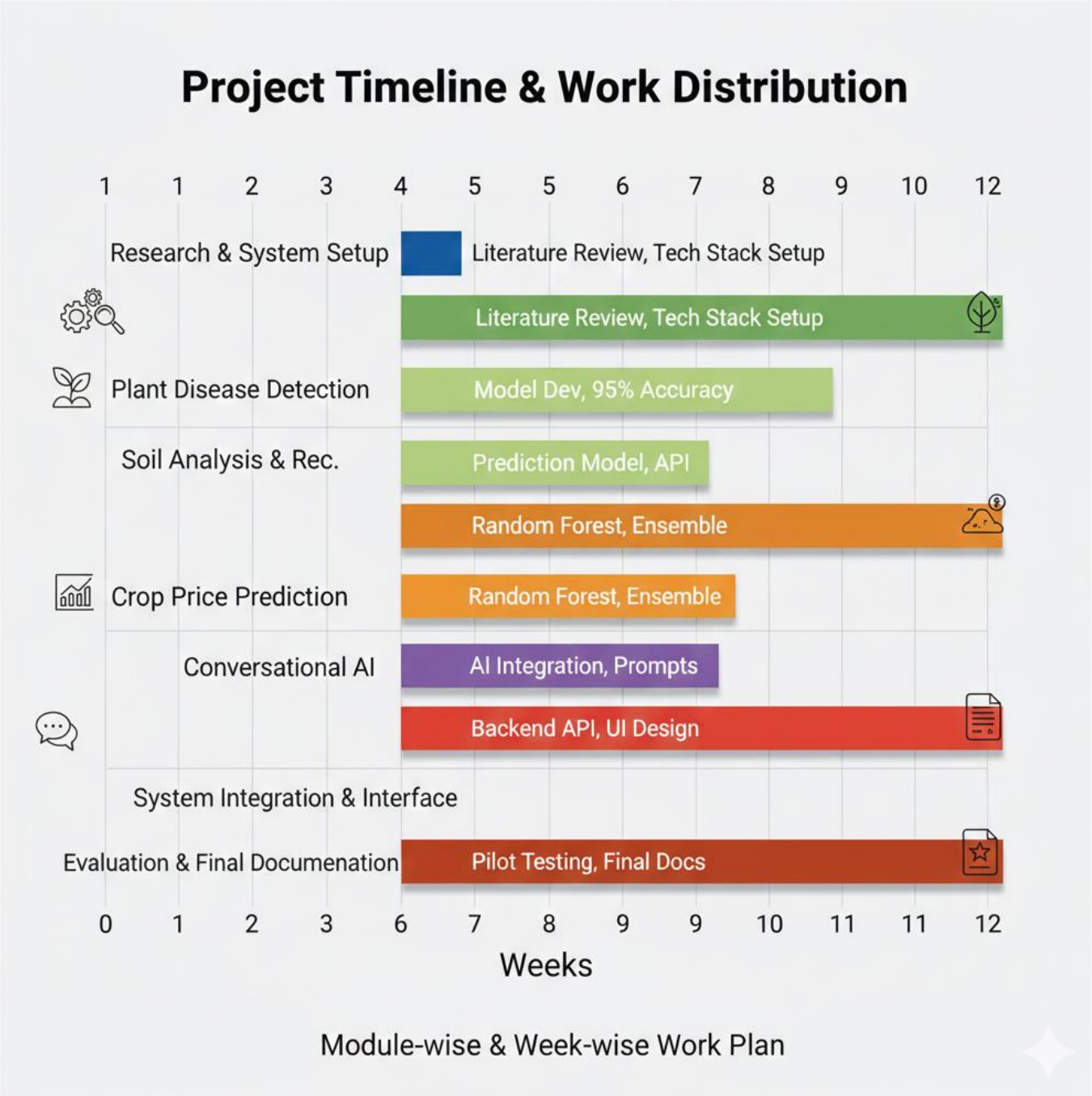
Overview

This module acts as the **integration layer** of the system, connecting all AI components through RESTful APIs, a web-based interface, and centralized data management. It enables seamless interaction between disease detection, soil analysis, price prediction, and the RAG-powered chatbot.

Key Features

- Unified REST API for all system modules
- Web-based and mobile-friendly user interface
- User authentication and session management
- Storage of analysis and chat history
- Model retraining endpoints for adaptive learning
- Basic analytics and monitoring dashboard

Work Plan:



Project Timeline and Milestone Schedule

Project Timeline (12 Weeks)

- Weeks 1–2: Research & System Setup

  Literature review, dataset collection (images, soil, price data), technology stack setup, system architecture design, database schema creation, and API specification.

- Weeks 3–4: Plant Disease Detection Module
  Image preprocessing pipeline, transfer learning–based model development, training and tuning to achieve 95%+ accuracy, disease knowledge database creation, and prediction API development.

- Weeks 5–6: Soil Analysis & Recommendation Module
  Soil quality prediction model, crop and fertilizer recommendation logic, database integration, API implementation, and validation testing.
- Weeks 7–8: Crop Price Prediction Module
  Feature engineering, Random Forest and Prophet model implementation, ensemble forecasting model development, and comparative evaluation against baselines.

- Week 9: Conversational AI Module
  Conversational AI integration, domain-optimized prompt templates, context management, and chat history storage.

- Weeks 10–11: System Integration & Interface
  Backend REST API integration, user authentication, frontend dashboard development, responsive UI design, and end-to-end system testing.

- Week 12: Evaluation & Final Documentation
  Pilot testing, performance evaluation, result analysis, research paper drafting, user manual preparation, and final presentation.

Work Distribution Chart

Work Distribution Chart

<u>Module Completion Timeline</u>

Week 1-2: [========Research & Setup========]

Week 3-4: [====Disease Detection Module====]

Week 5-6: [=====Soil Analysis Module=======]

Week 7-9: [===Price Prediction (Ensemble)==]

Week 10: [====Conversational AI Module====]

Week 11-13: [====System Integration==========]

Week 14: [====Pilot Testing==========]

Week 15: [====Comparative Analysis========]

Week 16: [====Final Documentation=========]

Project Impact:

Expected Impact

- **Farmer Benefits:** Faster decisions (60%), lower costs (20–30%), higher income (10–15%), and easy access to expert knowledge.
- **Agricultural Services:** Scalable, data-driven advisory with optimized use of extension resources.
- **Environmental Impact:** Reduced chemical use, improved soil health, and more sustainable farming practices.
- **Research Contribution:** Publishable AI frameworks, multi-domain integration, and reusable agricultural datasets.
- **Social Impact:** Improved food security, stronger farmer livelihoods, and increased rural technology adoption.

Future Enhancements:

 Expanded Disease Coverage:
Scale from 10 to 50+ diseases, including pest and weed detection.
 Multi-Crop Support:
Extend to 30+ crops with region-specific and horticulture recommendations.
 Weather Integration:
Real-time weather data with disease risk alerts and climate-based advice.
 IoT Integration:
Soil moisture and nutrient sensors enabling automated irrigation guidance.
 Multi-Modal AI:
Unified analysis of images, sensor data, text, and drone/video inputs.
 Federated Learning:
Privacy-preserving, region-aware model training across farms.
 Predictive Analytics:
Yield forecasting, outbreak early warnings, and long-term farm planning.
 Multi-Language & Voice Support:
Regional languages and voice interaction for inclusive farmer access.